Math-m-addicts group Y+: How to manipulate exponents.

Definition: For a real number a and a positive integer n, we define

$$a^n = a \cdot a \cdot a \ \ldots \cdot a \quad (n \ times).$$

Rule 1: For any real number a and positive integers m and n, we have:

$$a^{m+n} = a^m \cdot a^n.$$

Proof: This is clear as both sides evaluate multiplying a by itself $(m+n)$ times.

Rule 2: For any real number a and positive integers m and n, we have:

$$(a^m)^n = (a^n)^m = a^{m \cdot n}.$$

Proof:

$$(a^m)^n = a^m \cdot a^m \cdot a^m \cdot \ldots \cdot a^m = a^{m+m+\ldots+m} = a^{mn},$$

where multiplication is done n times and we used Rule 1 in the 3rd equality.

Example 1: What power of 4 is 8^{100} ?

Solution: $8^{100} = (2^3)^{100} = 2^{300} = (2^2)^{150} = 4^{150}$.

Underlying principle: All the subsequent definitions and conventions are made so that **Rule 1** and **Rule 2** stay valid.

Question 1: How should we define a^0 ?

Answer: We are free to define it in any way we please but, for it to be useful, we want the rules above to work. In particular, Rule 1 would imply that:

$a^m \cdot a^0 = a^{m+0} = a^m$ for any real number a and positive integer m.

Hence, if $a \neq 0,$ we must have $a^0 = 1$ if we have any hope to be able to extend Rule 1.

Definition: For any non-zero real number a, define $a^0 = 1.$

Question 2: For positive integer m, how should we define a^{-m} ?

Answer: We want the rules to work. In particular, Rule 1 would imply that:

$$a^{-m} \cdot a^m = a^{-m+m} = a^0 = 1 \text{ for any non-zero real number } a.$$

Hence, if we have any hope of extending the rules to negative powers, we must have:

$$a^{-m} = \frac{1}{a^m}.$$

Definition: For any non-zero real number a and positive integer m, define

$$a^{-m} = \frac{1}{a^m}.$$

Exercise 1: Find all solutions to the following equations:

a) $4^5 \cdot 2^x = 8^{10}.$
b) $4^3 \cdot 2^x = 1.$
c) $(\frac{1}{3})^{\frac{x}{2}} = 9.$
d) $(-\frac{1}{3})^{-3x} = 81.$
e) $(2^x)^x = 512.$

Exercise 2: Which of these numbers is larger?

a) 2^{100} or 5^{50} ?
b) 2^{100} or 10^{30} ?
c) 3^{200} or 10^{100} ?
d) 63^{30} or 33^{37} ?
e) 2^{105} or 5^{34} ?

Exercise 3: Find all integers x that satisfy the equations:

a) $(-x)^x = x^2.$
b) $(-x)^x = -x^2.$ (careful!).

Nested powers:

Frequently you encounter problems that have “nested powers”, i.e. an exponent which is itself a power of something. For example: $2^{2^{2^3}}$.

By convention, these are evaluated “top to bottom”. In particular:

$2^{2^{2^3}} = 2^{2^8} = 2^{256}.$

Example 2: Note that $2^{2^{100}} = 2^{(2^{99}\cdot 2)} = (2^{2^{99}})^2.$

Exercise 4: Circle the two expressions that evaluate to the same number:

$3^{3^{99+3}}$, $3^{3^{99}} + 3$, $3^{3^{99}\cdot 3}$, $3^{3^{99}} \cdot 3$, $3^{3^{99}+3}$, $(3^{3^{99}})^3$

Exercise 5: Using the previous example, prove that if $2^{2^{100}} - 1$ is fully factored, the factorization contains at least 100 (not necessarily distinct) prime factors.

Exercise 6:

a) Prove that if a, b are integers and n is a natural number, then $(a-b) \mid a^n - b^n$.
b) Prove that if number $2^n - 1$ is a prime, then n is also a prime.

Exercise 7: Using the digits 2, 2, 2 and 3 construct an expression that evaluates to as large of a number as you can, using addition, multiplication and exponentiation.